

Aqua Ride

Surf Point Management System

Group 03

Introduction

Surf schools and coastal rental hubs experience high volumes of seasonal traffic, necessitating a robust workflow for asset management, guest coordination, and scheduling. Manual tracking methods, unlinked spreadsheets, or paper logs easily introduce double-booking errors, inventory discrepancies, and data fragmentation. Managing individual customer accounts, matching skill levels to safety parameters, and assigning system privileges dynamically requires a unified, secure database application.

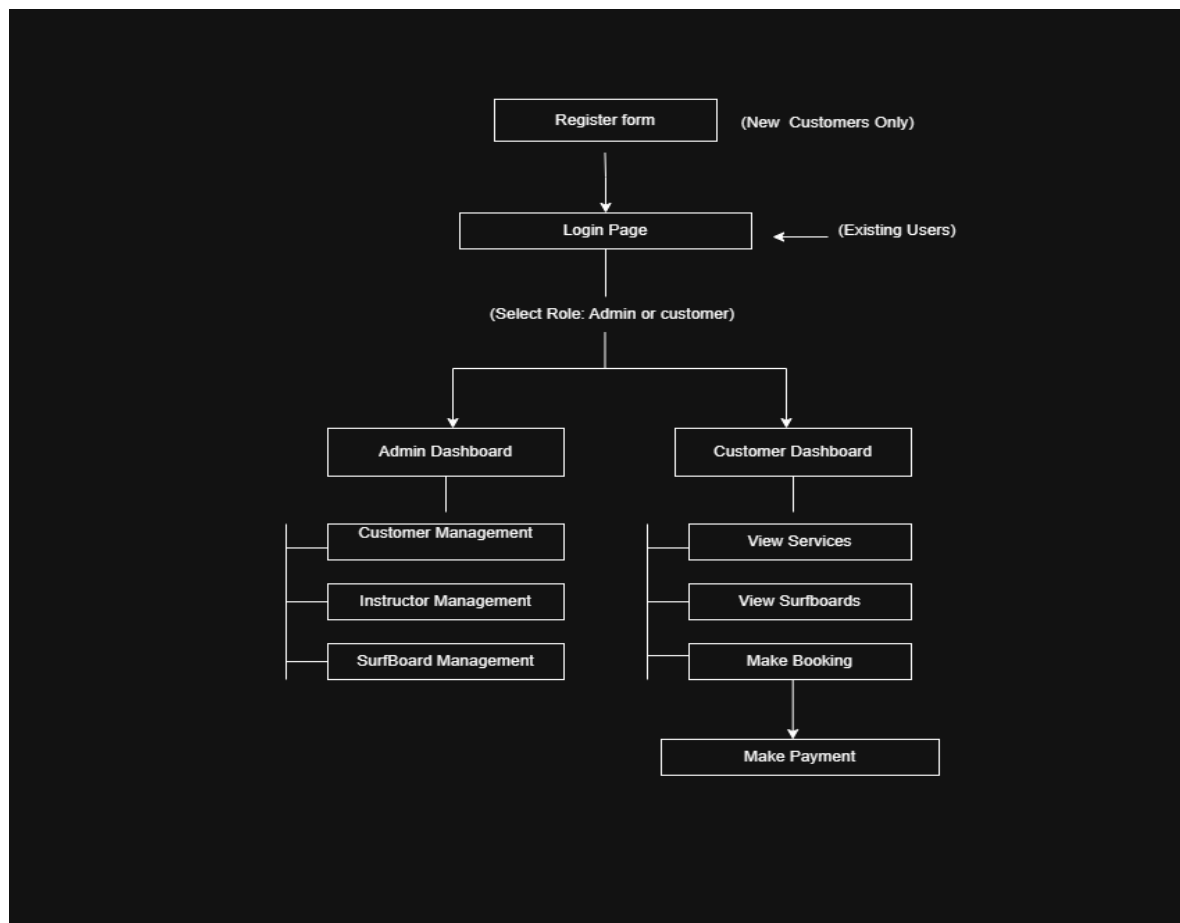
Project Aim & Objectives

The primary Aim of this project is to develop a desktop-based Surf Point Management System using *C#* and Microsoft SQL Server to automate and secure daily business operations.

To achieve this, the project focuses on the following core Objectives:

- **Secure Access Control:** Establish a role-based registration and login system for both administrators and customers.
- **Customer Data Management:** Maintain accurate client profiles, including contact details and surfing skill levels.
- **Inventory Tracking:** Manage surfboard stock levels and track rental availability in real time.
- **Booking Automation:** Streamline session reservations and digital logging to prevent scheduling conflicts.
- **Administrative Operations:** Provide central control panels for managers to update customer, instructor, and inventory records.

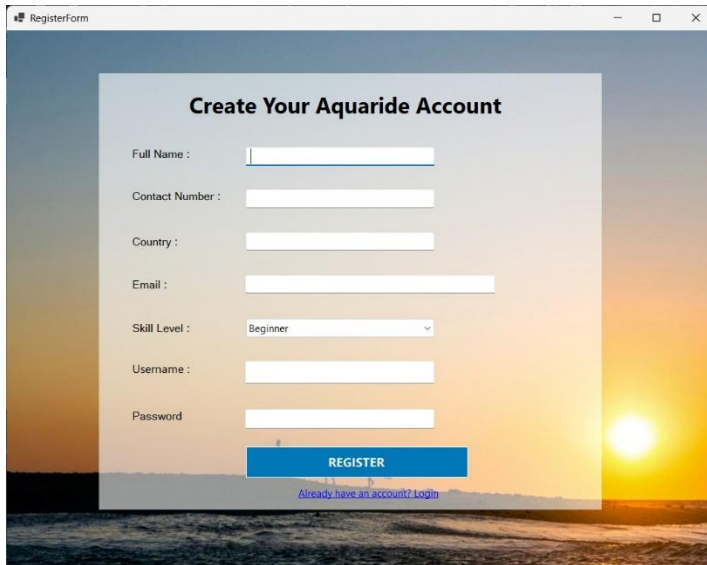
System Workflow & Main Functionalities



The Registration Gateway (Register Form)

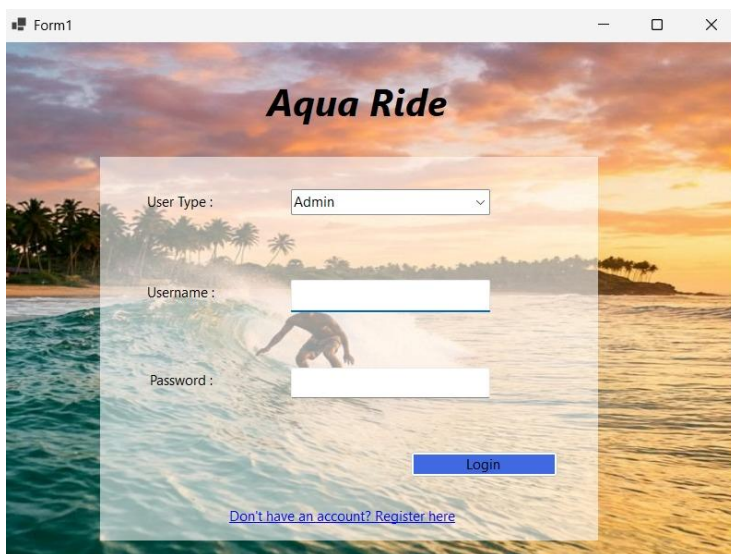
New clients must create an account first to interact with the system. The registration form captures extensive text and numerical fields, writing the data directly to the database.

- **Captured Data Fields:** Full Name, Contact Number, Country, Email, Skill Level (e.g., Beginner, Intermediate, Advanced), Username, and Password.

A screenshot of a web browser window titled 'RegisterForm'. The page has a background image of a sunset over a beach. In the center, there is a white box with the title 'Create Your Aquaride Account'. Below the title, there are several input fields: 'Full Name', 'Contact Number', 'Country', 'Email', 'Skill Level' (a dropdown menu with 'Beginner' selected), 'Username', and 'Password'. At the bottom of the form is a blue button labeled 'REGISTER'. Below the button, there is a link that says 'Already have an account? Login'.

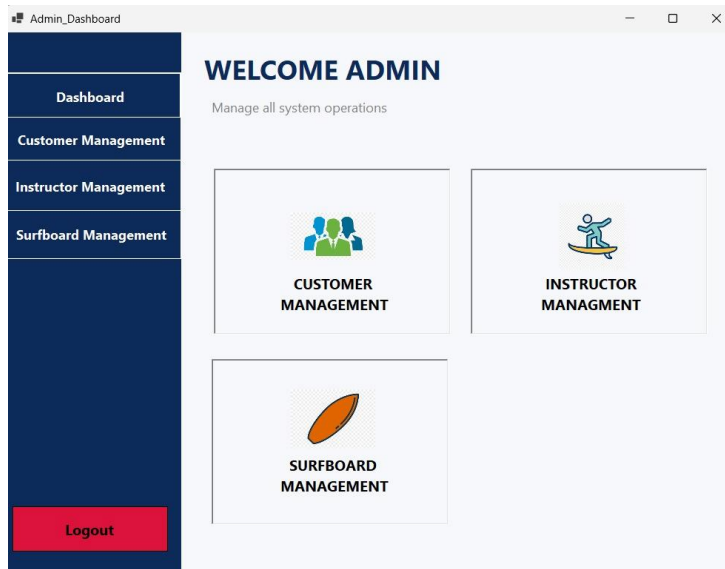
Secure User Entry (Login Page with Role Selection)

Existing users with established accounts skip registration and go directly to the Login page. To route users to their appropriate access levels, they must provide their credentials and explicitly choose their login tier via a selection option: **Customer** or **Admin**.

A screenshot of a web browser window titled 'Form1'. The page has a background image of a person surfing on a wave at sunset. In the center, there is a white box with the title 'Aqua Ride'. Below the title, there are three input fields: 'User Type' (a dropdown menu with 'Admin' selected), 'Username', and 'Password'. At the bottom of the form is a blue button labeled 'Login'. Below the button, there is a link that says 'Don't have an account? Register here'.

The Admin Dashboard (Data Management & Modification)

Choosing the **Admin** role validates the credentials against administrative credentials. Once logged in, the admin is granted central authority to manage and directly alter database tables across the system's three fundamental parties:



- **Customer Management Form:** Provides interfaces for the Admin to view and actively make changes to individual customer records directly within the system's backend **Customer Table**.

The screenshot shows the 'Customer Management' window. It has a light blue background and a title bar. The main area is titled 'CUSTOMER MANAGEMENT FORM'. It contains several input fields for 'Customer ID', 'Name', 'Contact Number', 'Email', 'Country', 'Skill Level' (a dropdown menu), 'User Name', and 'Password'. Below these fields are five buttons: 'Add', 'Update', 'Delete', 'Search', and 'Clear'. At the bottom, there is a table with the following data:

Customer ID	Full Name	User Name	Password	Contact Number	Email	Country	Skill Level
3	R.M.P.I Rathna...	PrasadiniN	Prasadini22	0726957230	pirathnayaka...	sri lanka	Intermediate
4	Kaveena	KaveenaNew	Kaveena2	0764567892	kaveena10@g...	India	Intermediate
5	Jayanai	Jr	Jr2	9658483087	hrhh@gmail...	india	Beginner
6	Pavani kavindya	Pavani	Pavani1	0785746367	hjrfdkm@gm...	america	Beginner
*							

- **Instructor Management Form:** Allows the Admin to monitor, update profiles, and make critical modifications to staff records housed inside the **Instructor Table**.

The screenshot shows a web application window titled "Instructor Management". It features a search bar at the top with the placeholder "Search Instructor ID:". Below this are input fields for "Instructor ID", "Experience Years", "Name", "Availability Status" (a dropdown menu), and "Contact Number". At the bottom of the form are five buttons: "Add", "Update", "Delete", "Search", and "Clear". Below the buttons is a table with the following data:

	Instructor ID	Name	Contact Number	Experience Years	Availability Status
▶	1	Akila	0768958493	5	Available
	2	kaveena	0694738392	1	Busy
*					

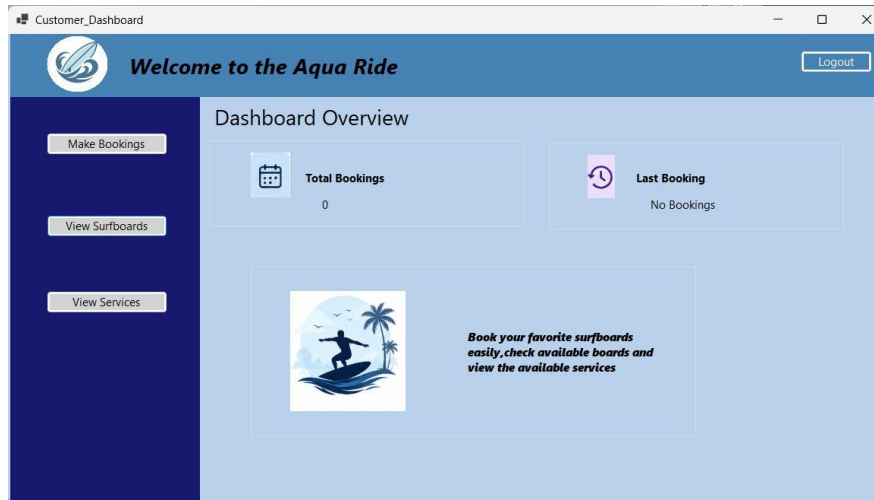
- **Surfboard Management Form:** Enables the Admin to alter technical entries, update availability flags, and save structural inventory updates inside the **Surf Board Table**.

The screenshot shows a web application window titled "Surfboard Management". It features a "Surfboard Details:" section with input fields for "Board ID", "Condition" (a dropdown menu with "New" selected), "Board Type" (a dropdown menu with "Shortboard" selected), "Price (Rs)", and "Status" (a dropdown menu with "Available" selected). Below this section are four buttons: "Add New", "Update", "Delete", and "Clear". Below the buttons is a table with the following data:

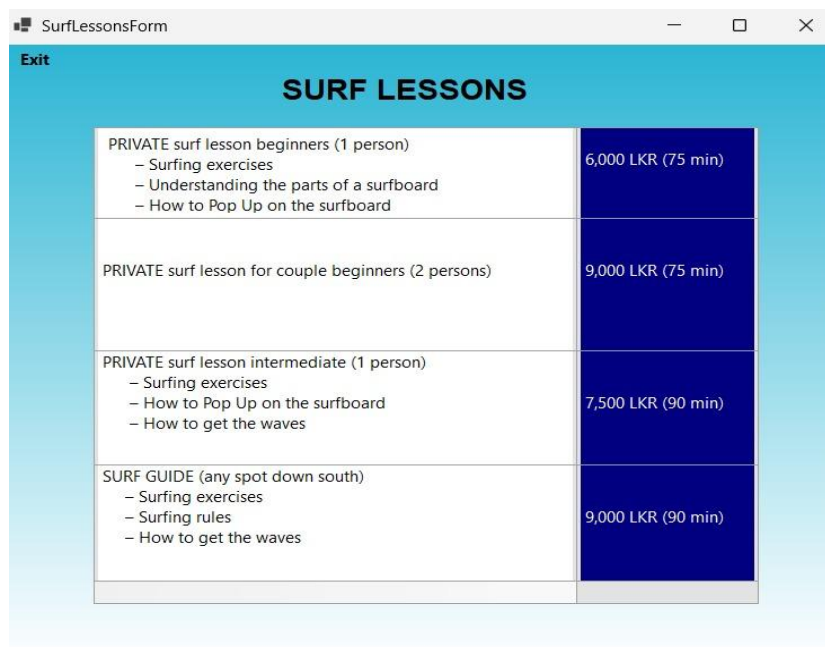
	BoardID	BoardType	ConditionType	Price	Status
▶	1	Longboard	New	3000.00	Available
	2	Shortboard	Good	2500.00	Available
	3	Fish	Fair	2800.00	Available
*					

The Customer Dashboard (Browsing & Transaction Gateway)

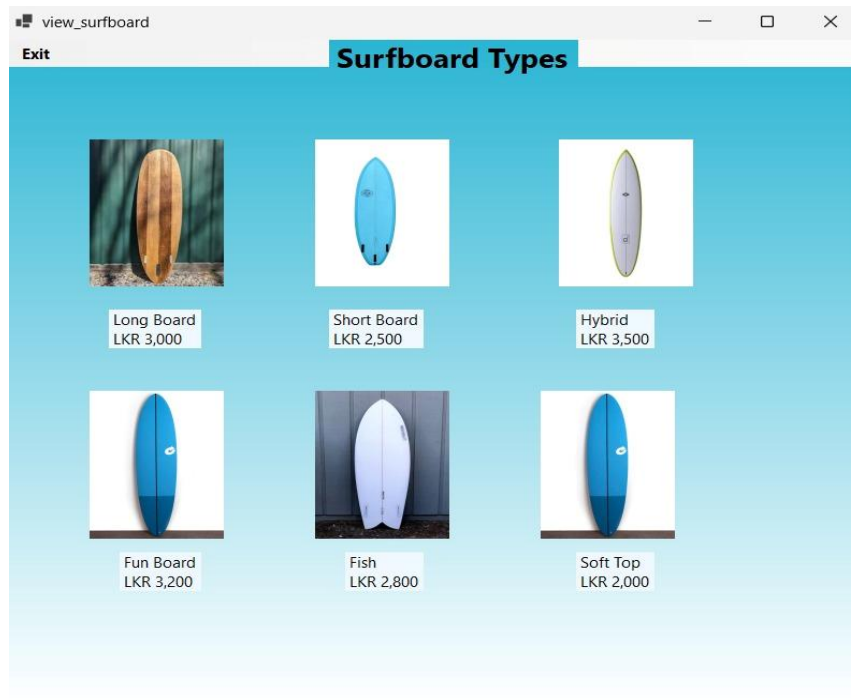
Choosing the **Customer** role routes validated clients to a custom dashboard tailored to booking utilities and system browsing. Customers cannot modify core backend tables directly; instead, their interface queries and interacts with the database through three dedicated forms:



- **View Services Form:** Dynamically loads and shows all the active surfing services, lesson packages, and operational programs currently offered within the system.



- **View Surfboard Form:** Displays a clear interface showing all the physical surf boards currently available in the system inventory for rental or use.



- **Make Booking Form:** An interactive gateway where a customer can input reservation details and directly place a session booking, generating a new transaction entry in the system.

The screenshot shows a web application window titled 'Booking_Form'. It features a header with 'File' and 'View', and a main title 'SURFBOARD BOOKING MANAGEMENT'. Below the header, there are input fields for 'Service', 'Surfboard Type', and 'Date'. To the right, there are fields for 'Booking ID', 'Customer ID', and 'Total Amount'. Below these fields are 'Submit' and 'Clear' buttons. At the bottom, there is a table showing a list of bookings:

BookingID	ServiceName	BoardID	BookingDate	TotalAmount
7	PRIVATE surf les...	2	6/21/2026	8500.00
8	PRIVATE surf les...	1	6/21/2026	9000.00
9	PRIVATE surf les...	1	6/10/2026	9000.00
10	PRIVATE surf les...	1	6/21/2026	12000.00
11	PRIVATE surf les...	2	6/21/2026	11500.00

Payment Processing & Billing

Automates checkout calculations by aggregating booking rates and hardware rental costs. This module processes transactions, tracks paid status, and records structural entries into the database.



The screenshot shows a web application window titled "Payment_Form" with a blue gradient background. The main heading is "PAYMENT MANAGEMENT". Below the heading, there are four input fields arranged in two rows. The first row contains "Booking ID" with a value of "12". The second row contains "Amount" with a value of "10000.00". The third row contains "Payment Method" with a dropdown arrow. The fourth row contains "Date" with a value of "6/21/2026" and a calendar icon. At the bottom, there are two buttons: "Pay Now" and "Clear".

Field	Value
Booking ID	12
Amount	10000.00
Payment Method	
Date	6/21/2026

Contribution of Each Group Member

Student Number	Member Name	System Responsibility	Forms & Modules developed
EC/2022/077	C.Niroshan	Core Identity Management & Main App Navigation	• Register Form (Fields: Name, Contact, Country, Email, Skill Level, Credentials) • Login Form (Role Selection logic) • Admin Dashboard
EC/2022/025	S.H.K.Chamarindi	Administrative Entity & Profile Management	• Customer Management Form (Handles updates to the Customer Table) • Instructor Management Form (Handles updates to the Instructor Table)
EC/2022/027	R.M.P.I.Rathnayake	Inventory & Catalog Modules	• Surfboard Management Form (Handles updates to the Surf Board Table) • View Surfboards Interface (Displays available surfboards to customers) • Customer Dashboard • Core Database Transaction Handling
EC/2022/029	H.M.J.Randima	Operational Services & Transaction Logic	• View Services Form (Displays active system services) • Make Booking Form (Processes customer reservations) • Payment Form (Handles financial updates to the Payment Table)